

Appendix 1: Full Table of Scenario A

Algorithm	Avg, Subopt. Arms	Avg. Regret	Avg. Zeros Count	Avg. Ones Count	Reward Variance	Subopt. Arm Ratio	p-value
UCB-Improved ($\delta = 1$)	325001.06	32500.11	132486.99	867513.01	2.143279e+09	0.32500	0.0
Greedy ($\varepsilon = 0.5$)	250022.03	25002.20	124979.36	875020.64	9.949043e+04	0.25002	0.0
ETC ($m = 100,000$)	100000.00	10000.00	110002.40	889997.60	8.343869e+04	0.10000	1.0
ETC ($m = 10$)	70040.82	7004.08	106989.53	893010.47	6.586476e+08	0.07004	0.0
Greedy ($\varepsilon = 0.1$)	50084.76	5008.48	104998.69	895001.31	8.004878e+04	0.05008	1.0
Greedy ($\varepsilon = 0.05$)	25184.69	2518.47	102516.59	897483.41	8.456372e+04	0.02518	1.0
ETC ($m = 10,000$)	10000.00	1000.00	100996.66	899003.34	7.795299e+04	0.01000	1.0
Greedy ($\varepsilon = 0.01$)	5880.14	588.01	100577.62	899422.38	1.073638e+05	0.00588	0.0
Greedy ($\varepsilon = 0.005$)	4229.88	422.99	100418.57	899581.43	1.066075e+05	0.00423	0.0
UCB	2381.75	238.17	100233.81	899766.19	7.851543e+04	0.00238	1.0
UCB-V ($\theta = 1, c = 1, b = 1$)	1099.26	109.93	100101.55	899898.45	7.868964e+04	0.00110	1.0
ETC ($m = 1,000$)	1000.00	100.00	100092.71	899907.29	7.838597e+04	0.00100	1.0
PAC-UCB ($c = 1, b = 1, q = 1.3, \beta = 0.05$)	994.99	99.50	100092.15	899907.85	7.923296e+04	0.00099	1.0
EUCBV ($\rho = 0.5$)	477.27	47.73	100038.73	899961.27	7.918856e+04	0.00048	1.0
UCB-Tuned	306.69	30.67	100022.63	899977.37	7.899541e+04	0.00031	1.0
ETC ($m = 100$)	100.95	10.10	100002.02	899997.98	7.884493e+04	0.00010	1.0

Tab. 1: Full Table of Scenario A – Summary Statistics across 100 Runs ($T = 10^6$)

Appendix 2: Full Table of Scenario B

Algorithm	Avg, Subopt. Arms	Avg. Regret	Avg. Zeros Count	Avg. Ones Count	Reward Variance	Subopt. Arm Ratio	p-value
UCB-Improved ($\delta = 1$)	505000.00	2525.00	102523.12	897476.88	6194415.18	0.50500	0.0
ETC ($m = 10$)	430160.03	2150.80	102151.03	897848.97	6432726.05	0.43016	0.0
ETC ($m = 100$)	410103.96	2050.52	102054.67	897945.33	6280731.62	0.41010	0.0
Greedy ($\varepsilon = 0.005$)	285740.62	1428.70	101425.39	898574.61	3665457.69	0.28574	0.0
ETC ($m = 1,000$)	258225.42	1291.13	101287.74	898712.26	4837381.73	0.25823	0.0
Greedy ($\varepsilon = 0.5$)	252903.26	1264.52	101262.24	898737.76	77552.71	0.25290	1.0
UCB	225434.51	1127.17	101119.98	898880.02	91872.83	0.22543	0.0
Greedy ($\varepsilon = 0.01$)	220636.24	1103.18	101092.95	898907.05	2434165.12	0.22064	0.0
ETC ($m = 100,000$)	100000.00	500.00	100491.10	899508.90	79356.56	0.10000	1.0
EUCBV ($\rho = 0.5$)	92272.12	461.36	100453.65	899546.35	109142.90	0.09227	0.0
PAC-UCB ($c = 1, b = 1, q = 1.3, \beta = 0.05$)	79179.44	395.89	100384.16	899615.84	94398.66	0.07918	0.0
Greedy ($\varepsilon = 0.05$)	72393.39	361.96	100352.51	899647.49	631969.28	0.07239	0.0
UCB-V ($\theta = 1, c = 1, b = 1$)	69317.27	346.58	100338.91	899661.09	96369.60	0.06932	0.0
Greedy ($\varepsilon = 0.1$)	68288.61	341.44	100335.52	899664.48	98984.19	0.06829	0.0
UCB-Tuned	42442.15	212.21	100204.46	899795.54	86826.78	0.04244	1.0
ETC ($m = 10,000$)	30132.64	150.66	100143.54	899856.46	510772.05	0.03013	0.0

Tab. 2: Full Table of Scenario B – Summary Statistics across 100 Runs ($T = 10^6$)

Appendix 3: Full Table of Scenario C

Algorithm	Avg, Subopt. Arms	Avg. Regret	Avg. Zeros Count	Avg. Ones Count	Reward Variance	Subopt. Arm Ratio	p-value
UCB-Improved ($\delta = 1$)	490000.14	2450.00	107456.68	892543.32	6169245.59	0.49000	0.00
ETC ($m = 10$)	429999.16	2150.00	107153.97	892846.03	6453855.77	0.43000	0.00
ETC ($m = 100$)	400123.02	2000.62	107008.36	892991.64	6217677.79	0.40012	0.00
Greedy ($\varepsilon = 0.005$)	352517.89	1762.59	106774.64	893225.36	4012222.43	0.35252	0.00
Greedy ($\varepsilon = 0.5$)	253809.30	1269.05	106277.85	893722.15	82824.39	0.25381	1.00
Greedy ($\varepsilon = 0.01$)	240771.05	1203.85	106212.53	893787.47	2644339.00	0.24077	0.00
UCB	234514.58	1172.57	106180.44	893819.56	93873.80	0.23451	0.78
ETC ($m = 1,000$)	231292.99	1156.46	106168.50	893831.50	4469433.28	0.23129	0.00
ETC ($m = 100,000$)	100000.00	500.00	105506.46	894493.54	77761.99	0.10000	1.00
EUCBV ($\rho = 0.5$)	95262.97	476.31	105485.88	894514.12	96233.97	0.09526	0.00
PAC-UCB ($c = 1, b = 1, q = 1.3, \beta = 0.05$)	83469.05	417.34	105425.50	894574.50	93935.53	0.08347	0.62
UCB-V ($\theta = 1, c = 1, b = 1$)	73529.84	367.65	105375.54	894624.46	90874.68	0.07353	1.00
Greedy ($\varepsilon = 0.1$)	72724.73	363.62	105374.35	894625.65	125787.02	0.07272	0.00
Greedy ($\varepsilon = 0.05$)	60310.19	301.55	105308.44	894691.56	177061.74	0.06031	0.00
UCB-Tuned	45217.90	226.09	105233.28	894766.72	81016.65	0.04522	1.00
ETC ($m = 10,000$)	34106.91	170.53	105179.43	894820.57	578944.69	0.03411	0.00

Tab. 3: Full Table of Scenario C – Summary Statistics across 100 Runs ($T = 10^6$)